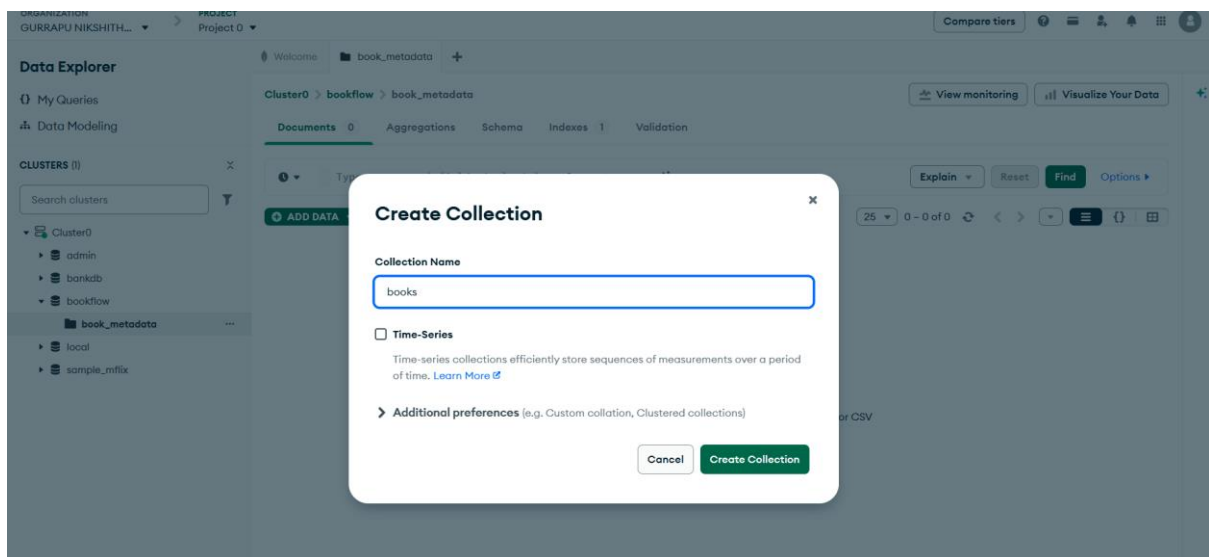
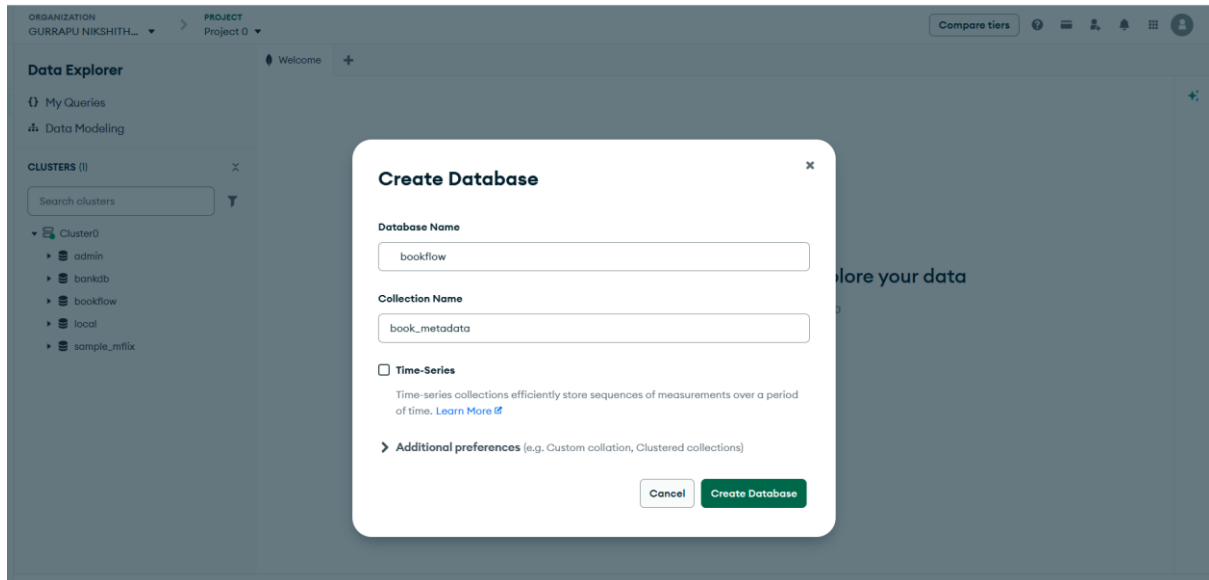


Week – 3

Gurrapu Nikshitha (2520030512)

Step 1 : Create Database & Collections



Step 2: Display & Insert the Documents into book_metadata

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

banknb

bookflow

book_metadata

books

local

sample_mfix

Welcome

book_metadata

books

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 1

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1-1 of 1

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

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books

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Welcome

book_metadata

books

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 2

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1-2 of 2

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Array (3)

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

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book_metadata

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+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 3

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1-3 of 3

book_id: 102

title: "Dune"

author: "Frank Herbert"

published_year: 2022

category: "Science Fiction"

format: "Digital EPUB"

file_size: "3.1 MB"

member_id: 102

reviews: Array (3)

book_id: 103

title: "Clean Code"

author: "Robert C. Martin"

published_year: 2019

category: "Programming"

format: "Physical"

dimensions: Object (3)

member_id: 103

reviews: Array (2)

ORGANIZATION

GURRAPU NIKSHITH...

PROJECT

Project 0

Compare tiers

Welcome

book_metadatadata

books

+

Cluster0

bookflow

book_metadatadata

View monitoring

Visualize Your Data

Documents 4

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25 1 - 4 of 4

_id: ObjectId('6a81e1d57393de16228ecc5a')

book_id: 183

title: "Clean Code"

author: "Robert C. Martin"

published_year: 2019

category: "Programming"

format: "Physical"

dimensions: Object (3)

member_id: 183

reviews: Array (2)

_id: ObjectId('6a81e1ef7393de16228ecc5c')

book_id: 184

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 184

reviews: Array (3)

Alert System Status: All Good

ORGANIZATION

GURRAPU NIKSHITH...

PROJECT

Project 0

Compare tiers

Welcome

book_metadatadata

books

+

Cluster0

bookflow

book_metadatadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25 1 - 5 of 5

_id: ObjectId('6a81e1ef7393de16228ecc5c')

book_id: 184

title: "Artificial Intelligence Basics"

author: "John Smith"

published_year: 2023

category: "Artificial Intelligence"

format: "Digital PDF"

file_size: "4.2 MB"

member_id: 184

reviews: Array (3)

_id: ObjectId('6a81e2117393de16228ecc5e')

book_id: 185

title: "Data Structures"

author: "Mark Allen"

published_year: 2024

category: "Computer Science"

format: "Digital PDF"

file_size: "5.0 MB"

member_id: 185

reviews: Array (2)

➤ Queries:

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
- book_metadata
 - books
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 1Validation

{
 "reviews.rating": {
 "\$gt": 4
 }
}

Generate query +

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

251 - 5 of 5

book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a81e1897393de1...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a81e1bc7393de1...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a81e1d57393de1...')	103	"Clean Code"	"Robert C. Martin"	2019	
4	ObjectId('6a81e1ef7393de1...')	104	"Artificial Intelligence ..."	"John Smith"	2023	
5	ObjectId('6a81e2117393de1...')	105	"Data Structures"	"Mark Allen"	2024	

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

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Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 1Validation

{
 "reviews.rating": {
 "\$in": [4, 5]
 }
}

Generate query +

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

251 - 5 of 5

book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a81e1897393de1...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a81e1bc7393de1...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a81e1d57393de1...')	103	"Clean Code"	"Robert C. Martin"	2019	
4	ObjectId('6a81e1ef7393de1...')	104	"Artificial Intelligence ..."	"John Smith"	2023	
5	ObjectId('6a81e2117393de1...')	105	"Data Structures"	"Mark Allen"	2024	

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

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Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 1Validation

{
 "format": {
 "\$regex": "Digital",
 "\$options": "i"
 }
}

Generate query +

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

251 - 4 of 4

book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a81e1897393de1...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a81e1bc7393de1...')	102	"Dune"	"Frank Herbert"	2022	
3	ObjectId('6a81e1ef7393de1...')	104	"Artificial Intelligence ..."	"John Smith"	2023	
4	ObjectId('6a81e2117393de1...')	105	"Data Structures"	"Mark Allen"	2024	

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

CLUSTERS (1)

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 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 2Validation

{ \$text: { \$search: "excellent" } }

Generate query +

Explain

Reset

Find

Options

ADD DATA +BULKEXPORT CODE

251 - 5 of 5

book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a81e1897393de1...')	101	"The Martian"	"Andy Weir"	2021	
2	ObjectId('6a81e1d57393de1...')	103	"Clean Code"	"Robert C. Martin"	2019	
3	ObjectId('6a81e1bc7393de1...')	102	"Dune"	"Frank Herbert"	2022	
4	ObjectId('6a81e2117393de1...')	105	"Data Structures"	"Mark Allen"	2024	
5	ObjectId('6a81e1ef7393de1...')	104	"Artificial Intelligence ..."	"John Smith"	2023	

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

CLUSTERS (1)

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book_metadata

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 2Validation

{ "\$text": { "\$search": "\"science fiction\"" } }

Generate query +

Explain

Reset

Find

Options

ADD DATA +BULKEXPORT CODE

251 - 2 of 2

book_metadata

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	
1	ObjectId('6a81e1bc7393de1...')	102	"Dune"	"Frank Herbert"	2022	
2	ObjectId('6a81e1897393de1...')	101	"The Martian"	"Andy Weir"	2021	

ORGANIZATION

PROJECT

Visual Tree

Raw Output

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

> TEXT_MATCH

Returned 2

Execution Time 0 ms

> FETCH

Returned 2

Execution Time 0 ms

> OR

Returned 2

Execution Time 0 ms

> IXSCAN

Returned 2

Execution Time 0 ms

Index Name: reviews.comments...

Multi Key Index: yes

> IXSCAN

Returned 2

Execution Time 0 ms

Index Name: reviews.comments...

Multi Key Index: yes

Query Performance Summary

2 documents returned

2 documents examined

0 ms execution time

Is not sorted in memory

4 index keys examined

Query used the following index:

_fts (text)

_ftsx

Close

➤ Normal Query :

The screenshot shows the Data Explorer interface with the 'book_metadata' cluster selected. The 'Documents' tab is active, displaying a list of 5 documents. A query is entered in the search bar: `{ "reviews.rating": { "$gt": 4 } }`. The results table shows the following data:

	_id ObjectId	book_id Int32	title String	author String	published_year Int32
1	ObjectId('6a81e1897393de1...')	101	"The Martian"	"Andy Weir"	2021
2	ObjectId('6a81e1bc7393de1...')	102	"Dune"	"Frank Herbert"	2022
3	ObjectId('6a81e1d57393de1...')	103	"Clean Code"	"Robert C. Martin"	2019
4	ObjectId('6a81e1ef7393de1...')	104	"Artificial Intelligence ..."	"John Smith"	2023
5	ObjectId('6a81e2117393de1...')	105	"Data Structures"	"Mark Allen"	2024

➤ Aggregations:

The screenshot shows the Data Explorer interface with the 'book_metadata' cluster selected. The 'Aggregations' tab is active, displaying a stage named '\$match'. The aggregation query is: `{ "published_year": { "$gt": 2020 } }`. The output preview shows 4 documents:

```
{ "_id": ObjectId('6a81e1897393de16228ecc56'), "book_id": 101, "title": "The Martian", "author": "Andy Weir", "published_year": 2021, "category": "Science Fiction", "format": "Digital PDF", "file_size": "2.5 MB", "member_id": 101 }
```

```
{ "_id": ObjectId('6a81e1bc7393de16228ecc57'), "book_id": 102, "title": "Dune", "author": "Frank Herbert", "published_year": 2022, "category": "Science Fiction", "format": "Digital EPUB", "file_size": "3.1 MB", "member_id": 102 }
```

ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
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- bookflow
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 - books
 - local
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book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

\$match

Sunwind

Generate aggregation

Explain

Run

Options

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 2 Sunwind

1 {

2 "path": "\$reviews"

3 }

Output preview after Sunwind stage (Sample of 10 documents)

_id: ObjectId('6a81e1897393de16228ecc56')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

_id: ObjectId('6a81e1897393de16228ecc56')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

Add stage

Alert System Status: All Good

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ORGANIZATION
GURRAPU NIKSHITH...

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

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- admin
- bankab
- bookflow
 - book_metadata
 - books
 - local
 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

\$match

Sunwind

\$sort

Generate aggregation

Explain

Run

Options

Untitled - modified

SAVE

CREATE NEW

EXPORT CODE

PREVIEW

STAGES

TEXT

WIZARD

Stage 3 \$sort

1 {

2 "average_rating": -1

3 }

4 }

Output preview after \$sort stage (Sample of 10 documents)

_id: ObjectId('6a81e1897393de16228ecc56')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

_id: ObjectId('6a81e1897393de16228ecc56')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

Add stage

Alert System Status: All Good

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Data Explorer

{} My Queries
⚙️ Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
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 - bookflow
 - book_metadata
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 - local
 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 2 Validation

Smatch Sunwind Sort Sgroup

Generate aggregation ⓘ Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD ⚙️

Stage 4 Sgroup

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after Sgroup stage (Sample of 4 documents)

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

```
_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3
```

Data Explorer

{} My Queries
⚙️ Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankab
 - bookflow
 - book_metadata
 - books
 - local
 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 2 Validation

Smatch Sunwind Sort Sgroup Edit

Explain Run Options

ALL RESULTS EXPORT CODE

Showing 1 - 4 count results < > ⌵ ⌶ ⚙️

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

```
_id: "The Martian"
average_rating: 4.666666666666667
total_reviews: 3
```

```
_id: "Artificial Intelligence Basics"
average_rating: 4.333333333333333
total_reviews: 3
```

```
_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3
```